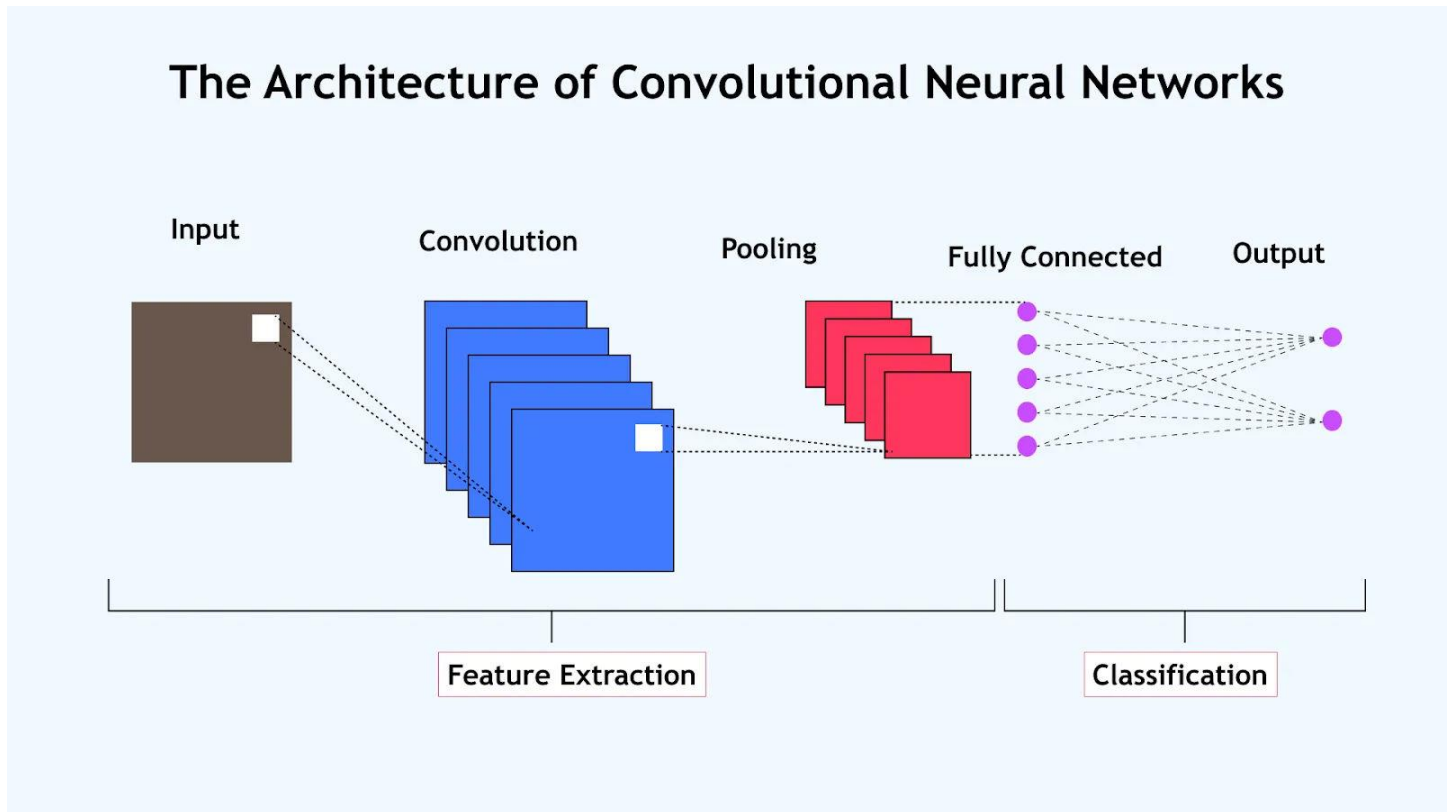


Module 4

Q. Convolutional Neural Network (CNN) Architecture

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Introduction

1. A Convolutional Neural Network (CNN) is a type of deep learning model mainly used for image and visual data processing.
2. CNN is inspired by the working of the human visual system.
3. CNN automatically learns important features from input data without manual feature extraction.
4. CNN is widely used in image recognition, object detection, and medical image analysis.
5. CNN contains multiple layers that process data step by step.

Main Components of CNN Architecture

1. Input Layer

1. The input layer receives the raw data such as an image.
2. The image is represented as a matrix of pixel values.
3. The input can be grayscale or color image.
4. Example: A 28×28 handwritten digit image is used in digit recognition.

2. Convolution Layer

1. The convolution layer is the main building block of CNN.
2. The layer uses filters or kernels to extract features from the input.
3. The filters slide over the input image to create feature maps.
4. The convolution operation detects edges, textures, and patterns.
5. Multiple filters produce multiple feature maps.
6. Example: Detecting edges of objects in an image uses convolution layers.

3. Activation Function (ReLU)

1. The activation function introduces non-linearity into the network.
2. ReLU is the most commonly used activation function in CNN.
3. ReLU converts negative values into zero.
4. ReLU helps the network learn complex patterns.

4. Pooling Layer

1. The pooling layer reduces the size of feature maps.
2. Pooling helps reduce computation and memory usage.
3. Pooling helps in extracting dominant features.
4. Max pooling is the most commonly used pooling method.
5. Example: Selecting the maximum value from a region of the image uses max pooling.

5. Fully Connected Layer

1. The fully connected layer connects all neurons from the previous layer.
2. The layer performs classification based on extracted features.
3. The layer works similar to a traditional neural network.
4. Example: Classifying an image as cat or dog uses fully connected layers.

6. Output Layer

1. The output layer produces the final prediction.
2. The Softmax activation function is commonly used for multi-class classification.
3. The output represents probabilities of different classes.

Working of CNN

1. The input image is passed to the convolution layer.
2. The convolution layer extracts features using filters.
3. The activation function introduces non-linearity.
4. The pooling layer reduces feature size.

5. The process repeats for multiple layers.
6. The fully connected layer performs classification.
7. The output layer gives final prediction.

Characteristics of CNN

1. CNN automatically learns features without manual effort.
2. CNN uses fewer parameters compared to fully connected networks.
3. CNN is highly effective for image data.
4. CNN captures spatial relationships in images.

Applications

1. CNN is used in face recognition systems.
2. CNN is used in self-driving cars for object detection.
3. CNN is used in medical imaging for disease detection.
4. CNN is used in handwriting recognition.

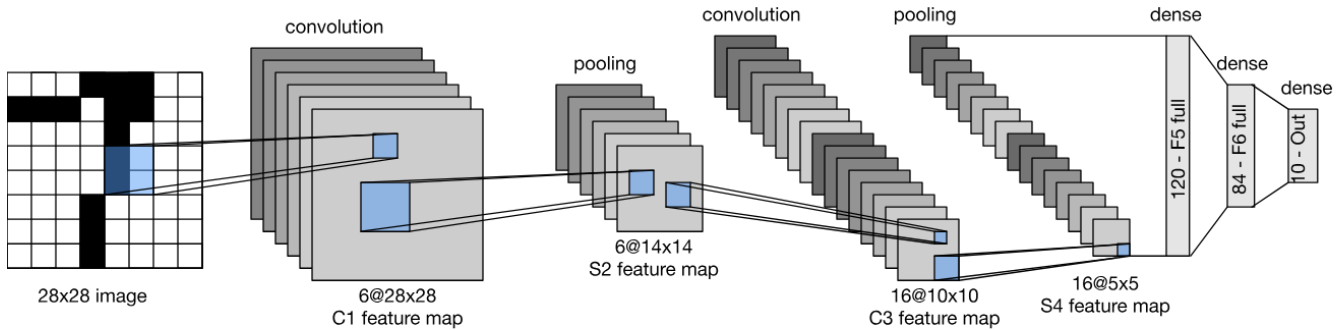
Q. Fully Connected Neural Network (FCNN) vs Convolutional Neural Network (CNN)

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Feature	Fully Connected Neural Network (FCNN)	Convolutional Neural Network (CNN)
Basic Structure	Every neuron is connected to all neurons in the next layer.	Neurons are connected only to local regions using filters.
Main Purpose	Used for general data like tabular data and simple classification.	Mainly used for image, video, and spatial data processing.
Feature Extraction	Features must be provided manually or learned inefficiently.	Features are automatically learned using convolution filters.
Number of Parameters	Large number of parameters because of full connections.	Fewer parameters due to weight sharing and local connections.
Computation Cost	Computationally expensive for large inputs.	More efficient for image data because of parameter sharing.
Spatial Information	Does not preserve spatial relationships between pixels.	Preserves spatial structure and relationships in data.
Layers Used	Dense layers are mainly used.	Convolution, pooling, and fully connected layers are used.
Overfitting	Higher chance of overfitting due to many parameters.	Lower risk of overfitting compared to FCNN for images.
Translation Invariance	Cannot recognize objects if position changes.	Can recognize objects even if position changes in image.
Input Type	Works well with numerical and structured data.	Works best with grid-like data such as images.
Training Speed	Training can be slower with large inputs.	Training is faster for image tasks due to fewer parameters.
Example Applications	House price prediction, medical data classification.	Face recognition, object detection, self-driving cars.

Q. LeNet

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Introduction

1. LeNet is one of the earliest Convolutional Neural Network (CNN) architectures.
2. LeNet was proposed by Yann LeCun in 1998.
3. LeNet was mainly designed for handwritten digit recognition.
4. LeNet successfully recognized digits in bank cheque processing systems.
5. LeNet is considered the foundation of modern CNN architectures.

Architecture of LeNet-5

LeNet-5 consists of seven layers excluding the input layer.

1. Input Layer

1. The input to LeNet is a grayscale image of size 32×32 pixels.
2. The input image contains handwritten digits from 0 to 9.
3. The input layer only passes the image to the next layer.

2. First Convolution Layer (C1)

1. The first convolution layer uses multiple filters to extract features.
2. The layer produces feature maps by applying convolution operation.
3. The output size becomes 28×28 feature maps.
4. The layer detects simple patterns like edges and curves.

3. First Pooling Layer (S2)

1. The pooling layer reduces the spatial size of feature maps.

2. The layer performs subsampling operation.
3. The output size becomes 14×14 feature maps.
4. Pooling helps reduce computation and overfitting.

4. Second Convolution Layer (C3)

1. The second convolution layer extracts more complex features.
2. The layer detects shapes and higher-level patterns.
3. The output feature maps become smaller but more meaningful.

5. Second Pooling Layer (S4)

1. The second pooling layer again reduces feature map size.
2. The layer improves computational efficiency.
3. The output becomes 5×5 feature maps.

6. Fully Connected Layer (C5 / F6)

1. The fully connected layer connects all neurons to perform classification.
2. The layer converts feature maps into a feature vector.
3. The network learns complex combinations of features.

7. Output Layer

1. The output layer contains 10 neurons representing digits 0 to 9.
2. The Softmax function is used to produce probabilities.
3. The neuron with highest probability gives the predicted digit.

Working of LeNet

1. The input image is passed through convolution layers to extract features.
2. Pooling layers reduce size and computation.
3. Fully connected layers perform classification.
4. The output layer predicts the digit class.

Characteristics of LeNet

1. LeNet uses convolution and pooling operations effectively.
2. LeNet has fewer parameters compared to fully connected networks.
3. LeNet is simple but powerful for small image datasets.
4. LeNet introduced concepts like feature maps and weight sharing.

Applications

1. LeNet is used for handwritten digit recognition.
2. LeNet is used in document recognition systems.
3. LeNet is used in bank cheque reading systems.
4. LeNet is used as a teaching model for CNN concepts.

Advantages

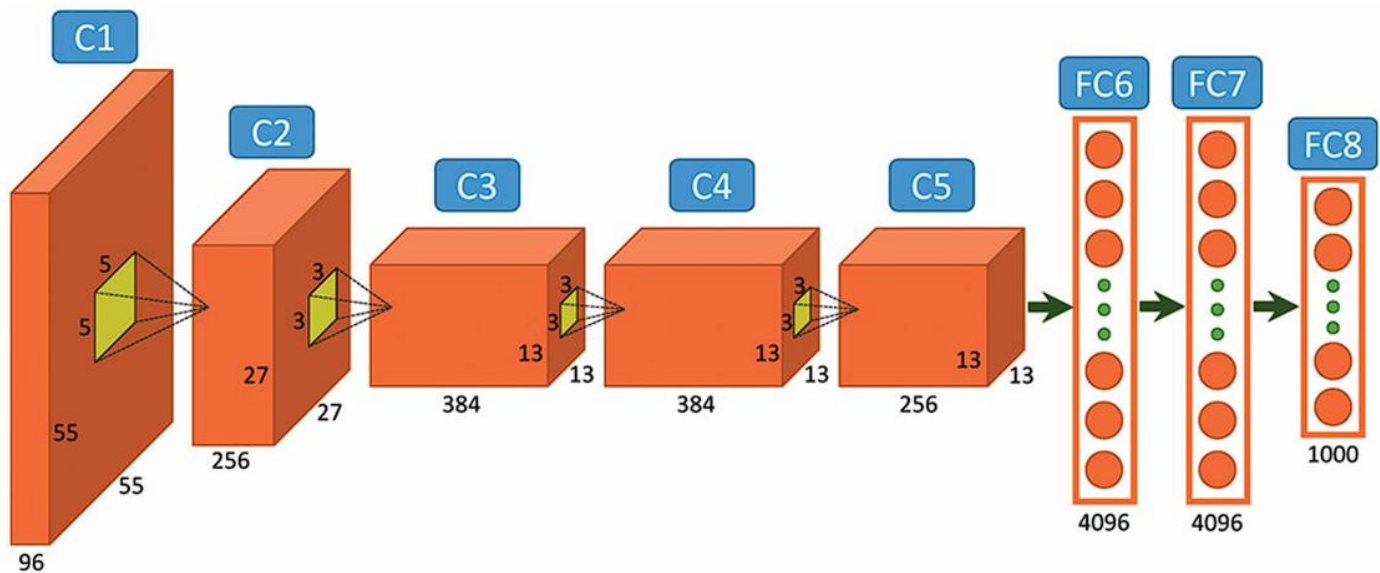
1. LeNet reduces computation using pooling layers.
2. LeNet automatically learns image features.
3. LeNet performs better than traditional machine learning for images.

Limitations

1. LeNet is not suitable for very large and complex images.
2. LeNet has limited depth compared to modern CNN architectures.
3. LeNet performance is lower than advanced models like ResNet.

Q. AlexNet

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Introduction

1. AlexNet is a deep Convolutional Neural Network proposed by Alex Krizhevsky in 2012.
2. AlexNet won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2012.
3. AlexNet achieved a very high accuracy compared to traditional methods.
4. AlexNet demonstrated the power of deep learning for image classification.
5. AlexNet is considered a breakthrough model in computer vision.

Architecture of AlexNet

AlexNet consists of 8 learned layers including 5 convolution layers and 3 fully connected layers.

1. Input Layer

1. The input image size used in AlexNet is $227 \times 227 \times 3$.
2. The input contains RGB color images.
3. The input layer passes image data to the convolution layer.

2. First Convolution Layer (Conv1)

1. The first convolution layer uses 96 filters of size 11×11 .
2. The stride used is 4 pixels.
3. The layer extracts basic features such as edges and textures.
4. The ReLU activation function is applied after convolution.

3. First Pooling Layer

1. The pooling layer uses max pooling operation.
2. The pooling reduces spatial size of feature maps.
3. Pooling improves computational efficiency.

4. Second Convolution Layer (Conv2)

1. The second convolution layer uses 256 filters of size 5×5 .
2. The layer extracts more complex patterns.
3. ReLU activation is applied after convolution.

5. Third, Fourth, and Fifth Convolution Layers (Conv3, Conv4, Conv5)

1. These layers use smaller filters of size 3×3 .
2. These layers learn deeper and more complex features.
3. ReLU activation is used after each convolution layer.
4. Pooling is applied after the fifth convolution layer.

6. Fully Connected Layers (FC6, FC7)

1. The fully connected layers contain 4096 neurons each.
2. These layers combine features learned from convolution layers.
3. Dropout is used to reduce overfitting.

7. Output Layer (FC8)

1. The output layer contains 1000 neurons for ImageNet classification.
2. The Softmax activation function is used for probability output.
3. The neuron with highest probability represents the predicted class.

Key Features of AlexNet

1. AlexNet introduced the use of ReLU activation instead of sigmoid or tanh.
2. AlexNet used dropout to reduce overfitting.
3. AlexNet used data augmentation to improve accuracy.
4. AlexNet was trained using GPUs for faster computation.
5. AlexNet used overlapping pooling to improve performance.

Working of AlexNet

1. The input image passes through multiple convolution layers to extract features.
2. Pooling layers reduce feature size and computation.
3. Fully connected layers perform classification.

4. The output layer produces final class prediction.

Applications

1. AlexNet is used for image classification tasks.
2. AlexNet is used in object recognition systems.
3. AlexNet is used in medical image analysis.
4. AlexNet is used in self-driving car vision systems.

Advantages

1. AlexNet significantly improved image classification accuracy.
2. AlexNet demonstrated the effectiveness of deep CNNs.
3. AlexNet can learn complex image features automatically.

Limitations

1. AlexNet requires high computational resources.
2. AlexNet has large number of parameters.
3. AlexNet is slower compared to modern lightweight architectures.